2D MODULAR GROUND KIT

Full Documentation

v1.0.0
Created in Unity 6.2

A free pixel-art environment kit for flexible 2D worlds

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1. Welcome

Thank you for downloading the 2D Modular Ground Kit. This free package is intended to provide a practical, flexible pixel-art foundation for 2D RPGs, cozy games, adventure games, prototypes, and other Tilemap-based projects.

The pack combines reusable terrain artwork with a sample scene and several optional runtime systems. You can use only the art, use the supplied Tile Palette, reuse the runtime systems, or use the sample scene as a reference for assembling your own environments.

Human-made assets: All included assets are human-made. The sprite artwork was created in GIMP, and the editable master GIMP source file is included so users can inspect and extend the artwork.

2. What Is Included

- Grass environment tiles
- Dry grassland environment tiles
- Snow / northern environment tiles
- Desert environment tiles
- Beach environment tiles
- Corrupted-land environment tiles
- Dirt paths and path pieces
- Water and shoreline tiles
- Natural borders and environmental detail sprites
- Props and decorative assets
- Unity Tile Palette
- Sample scene with a camera tour and ambient details
- Animated open-water helper
- Straight and corner shoreline wave systems
- Procedural biome blending system with same-Tilemap and cross-Tilemap modes
- Demo audio samples, ambient scripts, and prefabs
- Editable master spritesheet .xcf source file

3. Requirements and Compatibility

This release was created in Unity 6.2.

The supplied shoreline shaders target the Universal Render Pipeline 2D renderer and use Unity 2D shape-light support. The firefly demo can also use a URP Light2D component. For the supplied sample-scene look and the included lit shoreline effects, use a URP 2D project.

Art-only use: The sprite artwork and Tilemap content can be used independently of the optional shoreline, biome blend, and demo helper systems. If you only want the art, you do not need to use the runtime scripts.

4. Package Layout

```
2DModularGroundKit/
|-- Art/
|   `-- Tiles/
|       `-- Tile Palettes/
|-- Demo/
|   |-- Animations/
|   |-- Audio/
|   |-- Prefabs/
|   |-- Scenes/
|   `-- Scripts/
|-- Editor/
```

```
|-- Runtime/
|   |-- Materials/
|   |-- Scripts/
|   `-- Shaders/
`-- Documentation/
```

Runtime contains reusable environment functionality. Demo contains sample-scene-only helpers and presentation content. Editor contains editor-only tooling. This separation makes it easier to remove the demo while keeping the reusable parts of the kit.

5. Quick Start

1. Import the package into a Unity 6.2 URP 2D project.
2. Open the included sample scene under Demo/Scenes to see the intended visual setup.
3. Open Unity Tile Palette and select the supplied palette under Art/Tiles/Tile Palettes.
4. Create or select a Grid and Tilemap in your own scene, then paint the terrain pieces you want.
5. If you want animated open water, add PixelWaterTileAnimator to the water Tilemap and assign only the full-water sprites.
6. If you want shoreline waves, use the supplied shoreline materials and the appropriate shoreline controller.
7. If you want procedural transitions, create a BiomeBlendController2D with child BiomeBlendZone2D objects and rebuild the transitions.
8. Use the master .xcf file in GIMP if you want to create terrain combinations that are not pre-exported in the supplied spritesheets.

6. Artwork and Master GIMP Source

The package includes the master GIMP .xcf file used to assemble the main spritesheet. Grass, paths, terrain elements, and other source art are separated onto individual layers instead of being flattened into only the combinations supplied in the exported PNG.

This is intentionally included as a customization resource. You can turn layers on or off, combine different terrain elements, create additional variants, recolor artwork, and export your own sprites while keeping the original pack untouched.

Recommended workflow: Duplicate the master .xcf before making large changes. Keep the original package source intact, then export your customized PNGs under new names so Unity updates do not overwrite your edits.

6.1 Suggested Pixel-Art Import Settings

When importing your own edited exports, use the same pixel-art import approach as the supplied assets. In particular, avoid filtering that softens pixels and keep your pixels-per-unit choice consistent with the rest of your project.

The included demo camera tour uses a pixel-snapping value of 32 pixels per unit by default. If your project uses a different pixels-per-unit value, update that field on DemoCameraTour2D.

7. Tile Palette and Tilemap Workflow

The supplied Tile Palette is the fastest way to browse and paint the terrain. The kit is intentionally modular: combine terrain, paths, borders, water, and details using ordinary Unity Tilemap workflows.

- Keep major layers separated when useful, for example Ground, Water, straight Shorelines, and shoreline Corners.
- Separate Tilemaps make it easier to control sorting, materials, animation, and cross-Tilemap biome blending.
- The sample scene can be used as a reference for a working hierarchy.

8. Animated Open Water

PixelWaterTileAnimator animates cells that use one of the sprites listed in Water Sprites. The component belongs on the Tilemap containing full-water tiles.

Field	Purpose
Water Sprites	Sprites that identify full-water cells. Do not add shoreline sprites.
Frame Time	Time between animation states. Default: 0.25 seconds.

At startup, the script scans the Tilemap and records cells using the configured full-water sprites. It then animates all recorded cells in sync by changing each tile transform through four states: normal, horizontal flip, 180-degree rotation, and vertical flip.

Why this is useful: The animation is inexpensive and keeps the water texture itself simple. The procedural biome blending system also uses transparent overlays so the real animated water tiles can remain visible underneath transition pixels.

9. Shoreline Wave System

The shoreline system consists of separate straight-edge and corner controllers paired with custom URP 2D shaders. These systems create moving foam / crash motion and wet shoreline darkening while still participating in Unity 2D lighting.

9.1 Straight Shorelines

Use ShorelineWaveController2D with the supplied ShorelineCrash2D material / shader.

Setting	Description
Water Side	Which side of the shoreline faces the water:

	Up, Down, Left, or Right. AutoCorner exists in the enum but the dedicated corner system should normally be used for corner tiles.
Wave Speed	Controls the speed of the animated shoreline cycle.
Crash Distance	Controls the visual travel / reach of the crash effect.
Foam Strength	Controls foam visibility.
Wet Darkness	Controls how strongly wet shoreline pixels darken.
Phase Offset	Offsets the wave phase when variation is desired.

9.2 Corner Shorelines

Use ShorelineCornerWaveController2D with the supplied ShorelineCornerCrash2D material / shader. The controller examines nearby water cells and encodes the detected corner orientation into Tilemap color data so a single corner Tilemap can support all corner orientations.

Assign Water Tilemap directly whenever possible. If it is not assigned, the controller attempts to locate an object named Water beneath the same parent hierarchy.

- Detection Radius controls how far the corner detector checks for water.
- Detect Corner Directions can be run from the component context menu to refresh orientation metadata.
- The controller passes Grid cell pitch, including cell gap, to the corner shader so the effect remains aligned with the Tilemap.

9.3 Snow and URP 2D Lighting

Both shoreline shaders include special handling for snow-like pixels and expose snow threshold, snow wetness multiplier, and snow foam color properties. Internally, water detection uses both color saturation and blue dominance so bright neutral snow pixels are not treated as water.

The shaders use a Universal2D pass and Unity URP 2D shape-light support. This allows the custom shoreline result to respond to the same 2D lighting used by the rest of a URP 2D scene.

10. Procedural Biome Blending

The biome blending system creates irregular pixel-art transitions between terrain areas without requiring a dedicated transition sprite for every possible combination.

BiomeBlendController2D owns the generation process. Child BiomeBlendZone2D objects define where transitions occur, what direction they run, and which Tilemaps participate.

10.1 Controller Setup

9. Create an empty GameObject and add BiomeBlendController2D.
10. Add any reusable source Tilemaps to Source Tilemaps.
11. Create one or more child GameObjects and add BiomeBlendZone2D. A BoxCollider2D is required automatically.
12. Resize the zone in the Scene view to cover the area where you want the transition.
13. Choose Same Tilemap or Cross Tilemap mode and configure the zone.
14. Run Rebuild Biome Transitions on the controller.

10.2 Same Tilemap Mode

Use Same Tilemap when both terrain types are present on the same Tilemap. A zone can process all controller source Tilemaps or one Specific Tilemap.

10.3 Cross Tilemap Mode

Use Cross Tilemap when the two sides of the transition exist on different Tilemaps. A common example is Ground blending into Water.

15. Set Blend Source Mode to Cross Tilemap.
16. Assign Start Tilemap to the Tilemap representing the start side of the selected direction.
17. Assign End Tilemap to the Tilemap representing the end side.
18. Select Left To Right, Right To Left, Bottom To Top, or Top To Bottom.
19. Rebuild the transitions.

Zone Field	Description
Blend Direction	Defines how blend progression runs through the zone.
Irregularity	Blends between a simple midpoint threshold and procedural noise. Higher values produce stronger irregular breakup.
Noise Block Size	Controls procedural cluster size. 1 gives individual-pixel detail; larger values create chunkier clusters.
Seed	Changes the deterministic procedural pattern.
Zone Enabled	Enables or disables this zone.
Show Bounds	Shows the zone in Scene view.
Show Direction	Shows the blend direction arrow in Scene

	view.

10.4 Generated Overlay Behavior

Rebuild Biome Transitions first clears the prior generated result, then creates a __GeneratedBiomeBlend hierarchy beneath the controller. Generated SpriteRenderers are placed above the relevant Tilemap sorting order using the controller Sorting Order Offset.

The system generates transparent sprites that draw only transition pixels from the opposite biome. Pixels that belong to the real tile underneath remain transparent in the overlay. This is especially important for animated water because the underlying water tile remains visible and can continue animating.

Sprite pixels are read through a temporary RenderTexture. Source textures therefore do not need to be made Read/Write Enabled solely for this system.

Generated objects: Generated blend sprites and textures are runtime / editor-generated helper objects. Use Clear Biome Transitions before rebuilding manually if needed; Rebuild already performs a clear first.

11. Sample Scene

The sample scene is designed as both a visual showcase and a practical reference. It demonstrates several biome areas, water, shoreline animation, procedural transitions, ambient animation, camera-tour stops, and sample sound playback.

Demo-only scripts are intentionally stored separately from Runtime so you can remove presentation helpers without removing the reusable terrain systems.

12. Demo Helper Scripts

12.1 DemoCameraTour2D

DemoCameraTour2D requires a Camera. Each Camera Point can define a target Transform, travel time, pause time, optional orthographic zoom, optional stop sound, and stop-sound volume.

- Movement Curve controls easing between stops.
- Loop restarts the tour after the final stop.
- Pixel Snapping rounds camera X and Y to the configured pixels-per-unit grid. The supplied default is 32.
- If no AudioSource is assigned, the script reuses one on the Camera or creates one automatically. Stop sounds are played as one-shots when the camera reaches each stop.

12.2 CozyFirefly2D

CozyFirefly2D is a demo ambience component that requires a SpriteRenderer and TrailRenderer. It wanders around its starting position, adds vertical bobbing, pulses sprite brightness / alpha, can drive a URP Light2D, and builds a fading flickering trail.

> **Demo role:** This script is included as a reusable example, but it is not required to use the ground tiles, biome blender, water, or shoreline systems.

12.3 CozyTumbleweed2D

CozyTumbleweed2D moves an object continuously in a configurable 2D direction while rotating it around the Z axis. Move Speed, Move Direction, and Rotation Speed are exposed in the Inspector.

13. Demo Audio Samples

The Demo/Audio folder contains sample sound effects from Timbeaux's 2D Fantasy SFX Pack. That is a separate free Timbeaux asset pack. At the time the 2D Modular Ground Kit was prepared, the SFX pack was still pending Unity Asset Store approval.

These clips are included as sample ambience for the demonstration scene and for the camera-tour stop-sound feature, such as birds, crickets, frogs, owl calls, waves, and wind.

14. Technical File Reference

File	Role
Runtime/Scripts/BiomeBlendController2D.cs	Generates and clears procedural biome transition overlays; supports same-Tilemap and cross-Tilemap workflows.
Runtime/Scripts/BiomeBlendZone2D.cs	Defines blend bounds, source mode, Tilemap assignments, direction, irregularity, noise block size, seed, and Scene-view gizmos.
Editor/BiomeBlendZone2DEditor.cs	Editor-only Scene-view helper for resizing blend-zone bounds.
Runtime/Scripts/PixelWaterTileAnimator.cs	Finds configured full-water cells and animates their transform states in sync.
Runtime/Scripts/ShorelineWaveController2D.cs	Feeds straight shoreline orientation and wave settings into the straight shoreline material.
Runtime/Scripts/ ShorelineCornerWaveController2D.cs	Detects corner water direction, stores orientation metadata, and feeds wave / cell-

	pitch settings to the corner material.
Runtime/Shaders/ShorelineCrash2D.shader	URP 2D lit straight-shoreline wave / foam / wetness shader.
Runtime/Shaders/ ShorelineCornerCrash2D.shader	URP 2D lit corner-shoreline wave / foam / wetness shader with cell-pitch support.
Demo/Scripts/DemoCameraTour2D.cs	Looping camera-tour presentation helper with easing, zoom, pixel snapping, and per-stop audio.
Demo/Scripts/CozyFirefly2D.cs	Demo firefly movement, pulsing glow, Light2D, and trail helper.
Demo/Scripts/CozyTumbleweed2D.cs	Demo tumbleweed movement and full rotation helper.

15. Troubleshooting

Problem	Check
Water is not animating	Confirm PixelWaterTileAnimator is on the correct Tilemap and that Water Sprites contains the actual full-water sprite or sprites used by those cells.
Shoreline effect is missing	Confirm the TilemapRenderer uses the supplied shoreline material and that the appropriate shoreline controller is attached.
Straight shoreline moves the wrong way	Change Water Side on ShorelineWaveController2D.
Corner shoreline orientation is wrong	Assign the correct Water Tilemap and run Detect Corner Directions.
Shoreline ignores 2D lighting	Use the supplied URP 2D shoreline shaders and confirm the project is using a URP 2D renderer / lights.
Biome zone does nothing	Confirm Zone Enabled, confirm the zone covers the desired cells, choose the correct

	Same Tilemap or Cross Tilemap mode, and rebuild.
Cross-Tilemap blend does not appear	Assign both Start Tilemap and End Tilemap and verify the blend direction matches which side each Tilemap occupies.
Generated blend looks stale after edits	Run Rebuild Biome Transitions. The rebuild clears the previous generated hierarchy first.
Custom art looks blurry	Use pixel-art import settings and avoid filtered sampling that softens pixels.

16. Credits and Asset Authenticity

2D Modular Ground Kit was created for the Timbeaux line of 2D Unity tools and assets. All included assets are human-made, and the sprite artwork was created in GIMP.

The editable master .xcf source file is intentionally included so users can inspect the layered artwork and create additional terrain combinations instead of being limited to the pre-exported variants.

Demo sound effects are samples from Timbeaux's 2D Fantasy SFX Pack, a separate free pack that was pending Unity Asset Store approval when this documentation was prepared.

17. v1.0.0 Notes

v1.0.0 focuses on a solid, usable free environment foundation: multiple biomes, paths, water, shorelines, props, a reusable Tile Palette, editable source art, procedural biome transitions, and a polished sample scene. Additional content and combinations may be expanded in future updates.

Thank you for using the 2D Modular Ground Kit.